The Trace as Memory

Bookkeeping of what comes back — the patterns L F C, the categorical loop,
and their relevance to Kasami / Dobbertin

L linearized trace F Frobenius C cyclotomic cosets ⊗ categorical trace

A picture book for the `Equation1` Lean / Mathlib formalisation

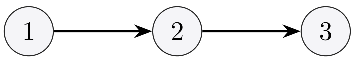
The thought that starts this note. *Maybe the trace is a sort of memory — bookkeeping, like an actual audit trail?* Yes. The trace *remembers only what returns to itself*: it discards every open path and keeps the closed loops — the feedback, the fixed points of R^k. It is bookkeeping of self-consistency: an audit of what comes back. In this project's case it records how each element cycles under Frobenius, summing it around its own orbit. This booklet turns that sentence into pictures, adds *variability* (several costumes of the same idea), draws the three patterns L F C, gives the categorical version ⊗, and pins down where each one touches Kasami / Dobbertin. Boxes marked ✓ correspond to `sorry`-free Lean declarations in `Equation1/`.

1. Trace = memory of round trips

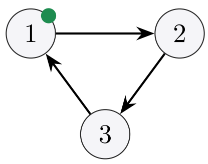
Think of a walker on a graph whose adjacency matrix is A (a relation drawn as arrows). The single most useful fact is:

$$\boxed{\mathrm{tr}(A^k) = \sum_i (A^k)_{ii} = \#\{\text{closed walks of length } k\} = \#\{x : R^k(x) = x\}}$$

The power A^k "takes k steps" (composes the relation k times, like iterating a function or counting successor-shifts in a `Nat`). The *trace* then throws away every walk that ends somewhere else and keeps only the ones that land home — the round trips. So k is *how many steps/shifts*, and the trace is *how many of those journeys are round trips.*



open path $1\to 2\to 3$
forgotten by tr

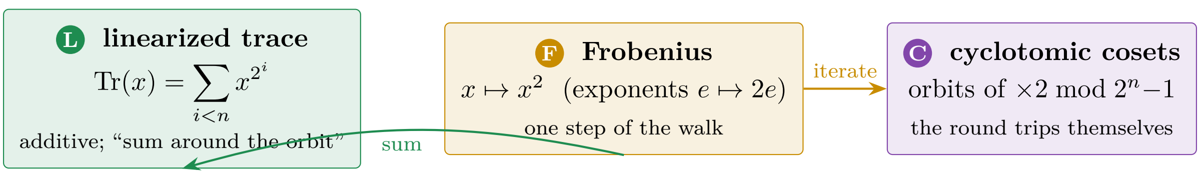


closed loop $1\to 2\to 3\to 1$
counted by $\mathrm{tr}(A^3)$

Two readings of the same A^k. *Boolean* ("does a k-step route exist?") gives the composed relation R^k; *linear* ("how many k-step routes?") counts walks. The trace only makes sense in the *linear* reading — which is exactly why it can *count* what comes back rather than merely detect it. The fixed-point twist: $\mathrm{tr}(A^k)$ picks out the x with $R^k(x) = x$, i.e. cycles whose length *divides* k. That last clause is the seed of the cyclotomic-coset pattern C below.

2. The three patterns, drawn — with variability

The formalisation is built from three recurring bricks. Each is a *different costume of the same round-trip idea*; the "variability" columns show the same brick appearing in several guises.

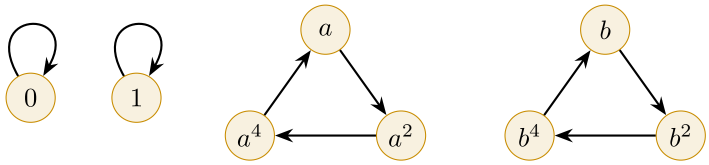


2.1 F Frobenius — one step of the walk

The Frobenius map $\sigma : x \mapsto x^2$ is a *bijective endorelation* on $L = \mathbb{F}_{2^n}$: a permutation, so its graph is a disjoint union of cycles. Iterating it k times is $x \mapsto x^{2^k}$ — "k shifts." The elements that come back after k steps, $\{x : x^{2^k} = x\}$, form the subfield $\mathbb{F}_{2^{\gcd(k,n)}}$, so

$$\operatorname{tr}(\sigma^k) = \#\{x : x^{2^k} = x\} = 2^{\gcd(k,n)}.$$

Here is the functional graph of σ on $\mathbb{F}_8$ $(n = 3)$: two fixed points and two 3-cycles.



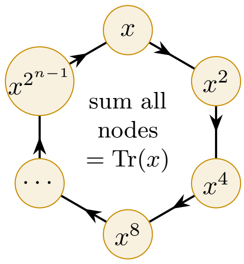
fixed: $\mathbb{F}_2$ two 3-cycles = cyclotomic cosets

Variability of F. Same brick, many costumes: (i) the field automorphism σ; (ii) doubling $e \mapsto 2e$ on exponents; (iii) the shift on cyclotomic cosets; (iv) periodicity $x^{p^r} = x^{p^{r \bmod n}}$. In Lean: `frob_cycle`, `frob_mod`, `frob_bijective`, `pow_field_bijective` ✓.

2.2 L Linearized trace — summing around the orbit

The absolute trace $\operatorname{Tr}(x) = \sum_{i<n} x^{2^i}$ is precisely "walk x around its own Frobenius orbit and add up what you see." It is $\mathbb{F}_2$-linear (additive), and it satisfies the Artin–Schreier *telescope* that glues L to F:

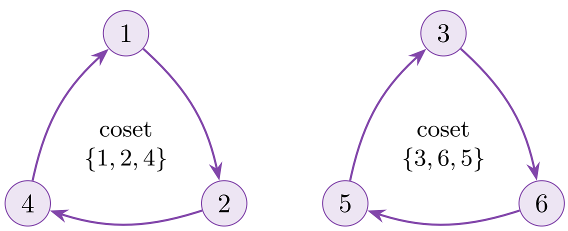
$$\boxed{L_m(x)^2 + L_m(x) = x^{2^m} + x} \qquad (\texttt{truncTrace\_sq\_add\_self})\ ✓$$



Variability of L. `Tr` (full, length n); `truncTrace` L_m (truncated, length m); additivity `truncTrace_add`; and — machine-checked in `Equation1/TraceLibrary.lean` — the identification with Mathlib's library trace: `Tr_eq_algebraMap_trace` shows $\operatorname{Tr} = \text{algebraMap} \circ \text{Algebra.trace}\,(\mathbb{F}_2)\,L$ ✓. So the "sum around the orbit" is literally the field trace.

2.3 C Cyclotomic cosets — the round trips themselves

The cycles of F *are* the cyclotomic cosets: orbits of $e \mapsto 2e$ on $\mathbb{Z}/(2^n-1)$. A cycle of length d contributes exactly when $d \mid k$ — this is "fixed points of R^k are cycles whose length divides k." In the equation-(1) extract this appears in its minimal, arithmetic form: coprimality and inverses mod $2^n - 1$.



$\times 2 \bmod 7$: two orbits of length 3 (plus the fixed 0)

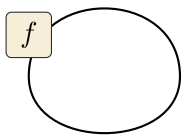
> **Variability of Ⓒ.** orbits of doubling; cyclotomic cosets mod $2^n - 1$; the divisibility rule "$d \mid k$"; and the arithmetic shadow used for Kasami: $\gcd(2^k-1, 2^n-1) = 1$ with the inverse $(2^k-1)^{-1} \bmod (2^n-1)$. In Lean: `mersenne_coprime`, `inv_mod_exists` ✓. (The full difference-set / 2-rank coset theory of the broader paper is *not* in this extract.)

3. The categorical pattern ⊗: one loop, drawn once

The reason all three costumes are "the same" is that trace is a single categorical gadget. In a monoidal category with duals, every object V has an evaluation $\varepsilon : V^* \otimes V \to I$ and coevaluation $\eta : I \to V \otimes V^*$. The trace of $f : V \to V$ is the one composite that *closes the wire into a loop*:

$$\mathrm{tr}(f) :\; I \xrightarrow{\eta} V \otimes V^* \xrightarrow{f \otimes \mathrm{id}} V \otimes V^* \xrightarrow{\text{swap}} V^* \otimes V \xrightarrow{\varepsilon} I.$$

ε (evaluation: close on top)



close the wire
into a loop
$\Rightarrow\ \mathrm{tr}(f) = \sum_i A_{ii}$

η (coevaluation: open at the bottom)

The same picture as a `tikz-cd` chain, with the k-th power inserted ("take k steps, then close the loop"):

$$I \xrightarrow{\quad\eta\quad} V \otimes V^* \xrightarrow{R^k \otimes \mathrm{id}} V \otimes V^* \xrightarrow{\quad\text{swap}\quad} V^* \otimes V \xrightarrow{\quad\varepsilon\quad} I$$

> **Why this is exactly the memory picture.** Coevaluation *opens* a path, R^k *walks* it k steps, and evaluation *closes* it — only paths that can be closed survive, i.e. the round trips. So $\mathrm{tr}(R^k)$ = feedback = fixed points of R^k. This is the same slogan as the *Lefschetz fixed-point theorem* (an alternating trace counts fixed points). Basis-independence $\mathrm{tr}(PAP^{-1}) = \mathrm{tr}(A)$ and $\mathrm{tr}(gf) = \mathrm{tr}(fg)$ fall out for free, because the loop never mentions a basis.

From the categorical loop to this project's trace. For $K \subseteq L$ and $x \in L$, multiplication $m_x(y) = x \cdot y$ is an endomorphism of L, and $\mathrm{Tr}_{L/K}(x) := \mathrm{tr}(m_x)$ is the categorical loop applied to m_x. Over $K = \mathbb{F}_2$ this unwinds (Mathlib: `FiniteField.algebraMap_trace_eq_sum_pow`) to $\sum_{i<n} x^{2^i}$ — the project's own Ⓛ. So ⊗ is not a fourth pattern: it is the hinge that makes Ⓛ = "sum around the Ⓕ-orbit" inevitable.

4. Relevance to Kasami / Dobbertin

Pattern	Memory / round-trip reading	Role in Kasami / Dobbertin (Eqn. (1))
Ⓕ Frobenius	one step of the walk; $\mathrm{tr}(\sigma^k) = 2^{\gcd(k,n)}$	the squaring that generates every cycle; drives (1) ⇒ (2) and both cases of Thm 1
Ⓛ Trace	sum around the orbit; audit of what returns	Tr states equation (1); the telescope $L_m(x)^2 + L_m(x) = x^{2^m} + x$ is Dobbertin's "add the 2^k-th power to itself" step
Ⓒ Cosets	the round trips themselves; length $d \mid k$	$\gcd(2^k-1, 2^n-1) = 1$ and the inverse mod $2^n - 1$ decide when the Kasami power $x^{2^{2k}-2^k+1}$ permutes
⊗ Categorical	the single loop behind all three	identifies the project's Tr with the library field trace, guaranteeing the bookkeeping is the standard one

Verdict. The memory / trace-of-powers picture *does* apply to the Dobbertin equation-(1) material: the trace L, the doubling F, and the coprime/coset bookkeeping C are exactly the moving parts of Theorem 1, and the categorical loop ⊗ certifies that the project's trace is the genuine field trace (`Tr_eq_algebraMap_trace`, `Equation1/TraceLibrary.lean`) ✓. Honest caveat: in this extract the "trace of powers" appears in its specialised *Frobenius-permutation* form ($\mathrm{tr}(\sigma^k) = 2^{\gcd(k,n)}$, cycles whose length divides k), not as a graph-triangle count; and the broader difference-set / 2-rank coset theory of the full paper is outside the extract.

Appendix. The conversation this note draws on

Q. Maybe trace is a sort of memory, or bookkeeping like an actual trace?

A. Yes — the trace "remembers" only what returns to itself: it discards all the open paths and keeps the closed loops (the feedback, the fixed points of R^k). So it is bookkeeping of self-consistency — an audit of what comes back. In this project's case it records how each element cycles under Frobenius, summing it around its own orbit.

Q. So it is a bit like how many times a walker goes around a torus / loop, or how in accumulator functions, or in inductive `Nat` a number carries a sense of how many steps / shifts?

A. Exactly that flavour. $\mathrm{tr}(A^k)$ counts "things that come back after k steps" — closed loops:

- *Torus / loop:* $\mathrm{tr}(A^k)$ counts the ways to walk out and land home in k steps — winding around and returning.
- *Accumulator / inductive-`Nat`:* composing R k times (R^k) *is* "take k steps," like iterating a function or counting shifts; the k in A^k is that step-count, the successor-depth of a `Nat`.
- *Fixed-point twist:* tr picks the feedback — points with $R^k(x) = x$, cycles whose length divides k. For Frobenius $x \mapsto x^2$ on $\mathbb{F}_{2^n}$, "k shifts" means squaring k times; the ones that return form a subfield and the cycles are the cyclotomic cosets.

So: k = how many steps / shifts, trace = how many of those journeys are round trips.

Companion to `PaperStory.tex`, `Equation1_LEGO.tex`, `TraceMap_Categorically.tex`, `TracePowers_Relations.tex`.
Rebuild: `tectonic TraceAsMemory.tex`.